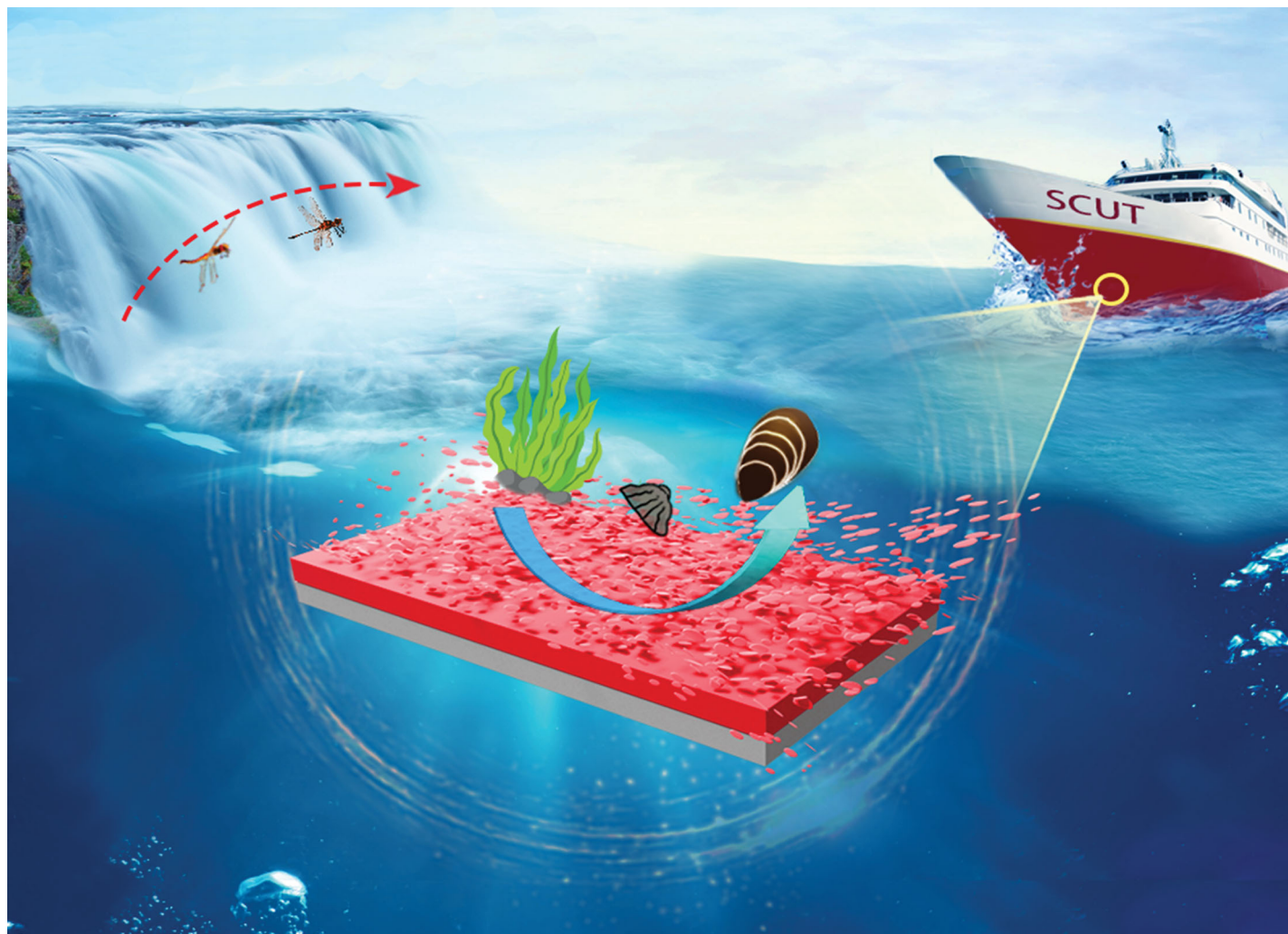




Highlighting research from the Marine Engineering Materials Team, led by Prof. Guangzhao Zhang and Prof. Chunfeng Ma at South China University of Technology.

Dynamic surface antifouling: mechanism and systems

Marine biofouling is a global problem. High efficiency and eco-friendly antifouling systems are in pressing need. Ma *et al.* proposed the concept of Dynamic Surface Antifouling (DSA), where the dynamic surface refers to a changing surface that continuously renews itself in seawater and thus decreases the adhesion of biofouling. Based on this strategy, they have developed a series of degradable polymer-based dynamic surfaces with tunable renewability, excellent antifouling and mechanical performance. Such a novel antifouling system is an effective and promising strategy.

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Dynamic surface antifouling: mechanism and systems

Qingyi Xie, Jiansen Pan, Chunfeng Ma * and Guangzhao Zhang *

Marine biofouling is a global problem today. High efficiency and eco-friendly antifouling systems are in pressing need. In recent years, we have proposed the concept of dynamic surface antifouling (DSA). That is, a continuously changing surface can effectively prevent marine fouling organisms from landing and adhesion. Based on this strategy, we developed coatings with dynamic surfaces by using degradable polymers including polyester-polyurethane, modified polyester and poly(ester-co-acrylate). They exhibit tunable renewability, and excellent antifouling and mechanical performance. Moreover, the polymers can serve as carrier and controlled release systems of antifoulants so that they have long service life. This paper reviews the progress and trends in marine anti-biofouling, and presents the mechanism and systems of DSA.

1 Introduction

Marine biofouling arising from the undesirable accumulation of marine microorganisms, plants and animals on submerged surfaces, has adverse impacts on marine industries and maritime activities.^{1–4} It increases the weight and roughness of ship hulls, which causes the frictional resistance to drastically increase and causes additional fuel consumption as well as emission of greenhouse gases. Biofouling initiates or accelerates the corrosion process of metal and concrete structures,⁵ raising safety concerns for cross-sea bridges and other facilities. It also blocks seawater pipelines in near-sea energy infrastructure

such as nuclear power plants and pumped-storage hydro-electricity stations, which decreases the heat exchange efficiency and affects the operation of generators.⁶ In the aquaculture industry, fouling organisms attached on cage netting hinder nutrient exchange and deform the cage due to extra weight, consequently lowering the production.⁷ Moreover, fouling organisms on vessels migrate to different seas where they do not naturally exist and disturb the ecological system.⁸ Therefore, marine anti-biofouling is economically and environmentally important.

It is generally recognized that the growth of marine biofouling usually includes several key steps (Fig. 1).⁹ Firstly, a conditioning film quickly forms on the submerged surface due to adsorption of proteins, glycoproteins and polysaccharides. Then, bacteria adhere on the conditioning film and gradually develop into a biofilm by secreting extracellular polymeric substances

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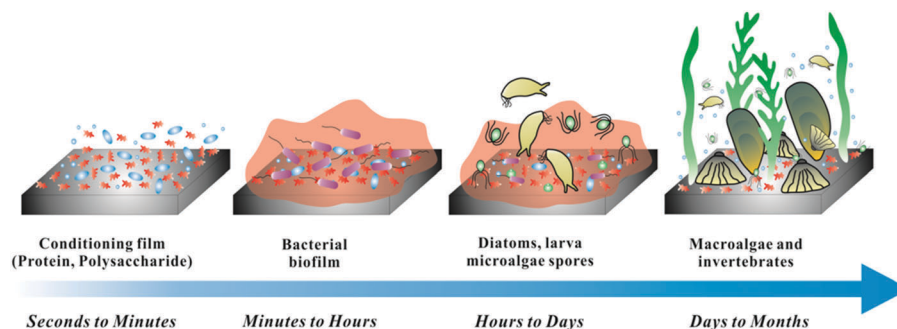


Fig. 1 The typical growing process of marine biofouling.

consisting of proteins and polysaccharides to envelope and anchor themselves. Subsequently, diatoms and macroalgae spores accumulate on the surface as the biofilm provides nutrients. Finally, larvae of marine macroorganisms such as barnacles settle and grow as macrofoulers. However, this successional model can only be applied to some marine organisms and many others behave differently. For instance, the cyprids of barnacle *Amphibalanus Amphitrite* can settle on a surface without the presence of a biofilm.¹⁰ Actually, there are more than 4000 fouling species in the sea and the marine environments vary considerably in different areas regarding temperature, salinity, pH, *etc.*,¹¹ making the dominant fouling organisms different as well. Moreover, laws and regulations are becoming increasingly strict with the use of antifouling materials that may influence the marine ecology. Under these complexities, it is now a worldwide challenge to develop eco-friendly, broad-spectrum, applicable antifouling technologies.

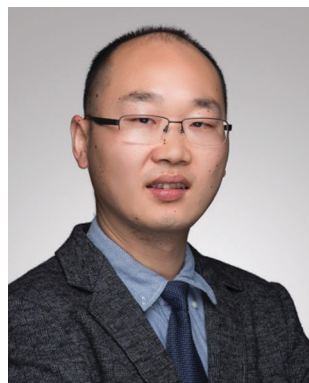
2 Current marine antifouling approaches

To date, several antifouling approaches have been developed to combat biofouling, including coatings, mechanical cleaning,¹²

electrolyzing seawater,¹³ ultraviolet (UV) irradiation,¹⁴ ultrasonic treatment¹⁵ and so on. Table 1 summarizes these methods regarding their main components, antifouling mechanisms and characteristics. Using antifouling coatings is the most effective, economic and widely used strategy among them. Self-polishing copolymer-based coatings containing tributyltin (TBT) have high efficiency in preventing the settlement and growth of marine organisms so they were once the best choice. However, they were globally banned in 2008 because of their persistent toxicity to non-target organisms.¹⁶ Since then, there has been a pressing need for developing environmentally friendly systems to prevent marine biofouling.

2.1 Antifoulant releasing coatings

Although the TBT based coating is no longer an option, coatings carrying other chemically active antifoulants (biocides) with comparatively lower toxicity are still widely used. Their performance is significantly influenced by polymer binders, *i.e.* the carriers and releasing matrixes, because binders determine the release of antifoulants.¹⁷ Certainly, other components in coatings like antifoulants and pigments also influence the properties. Currently there are three types of polymer binders (Fig. 2), including insoluble binders, controlled depletion



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in the South China University of Technology. His research interests cover marine antifouling materials, chemical grouting materials and physical chemistry of macromolecules.

Table 1 Current antifouling approaches

Approach	Main components	Mechanism	Remarks	Ref.
Antifoulant releasing coatings	Polymer binders (insoluble binders, CDP and SPC) and antifoulants	Release of toxic antifoulants from binders	Widely applied on ships; detrimental to marine ecology; declined efficacy in idle periods, even for SPC based coatings after 2–3 years	19–27
Fouling release coatings	Silicone; silicone containing amphiphilic block copolymers; fluorinated polymers	Non-stick properties due to low surface energy and low elastic modulus	Commercially available; capable of reducing drag; poor mechanical properties; low efficiency in idle periods after 2–3 years	1, 2 and 28–42
Biomimetic materials	Micro-structured surfaces	Lowering contact points for adhesion	Reducing adhesion of some organisms; not proven effective in marine environments; difficult to be applied	46–53
	Antifoulants	Inhibiting settlement and adhesion of marine organisms by natural antifoulants or enzymes	Natural antifoulants: difficult and expensive to obtain; enzymes: not broad-spectrum and hard to retain activity	55–61
	Slippery surfaces	Lubricating effect of oil additives	Reduce fouling adhesion but possibly declining ability with time; some are textured surfaces	52–66
	Deforming surfaces; sacrificial skins	Detaching biofouling by the deformation or renewal of the surface	Deforming surfaces: can be complementary to other methods; sacrificial skins: not durable because of the fast dissolution	67–69
Protein resistant polymers	Zwitterionic polymers; PEG based polymers; hydrogels	Hydration layer as a physical barrier for fouling organisms	Efficient in laboratorial bioassays but not in seawater for long periods; poor mechanical properties	70–100
Amphiphilic polymers	Copolymers of PEG and polystyrene, silicone or fluorinated polymers.	Possibly due to an “ambiguous” effect of both hydrophobic and hydrophilic domains	Mechanism not totally clear; efficient in indoor assays but few reports on properties in seawater; used as additive for FRCs	101–112
Others	Mechanical cleaning; ultrasound treatment; ultraviolet irradiation; electrifying seawater	Creating physically or chemically unsuitable conditions for organisms	Not easy to apply and restricted to certain occasions (e.g. electrifying seawater technique mainly used in ballast tank of vessels); supplementary to antifouling coatings	12–15

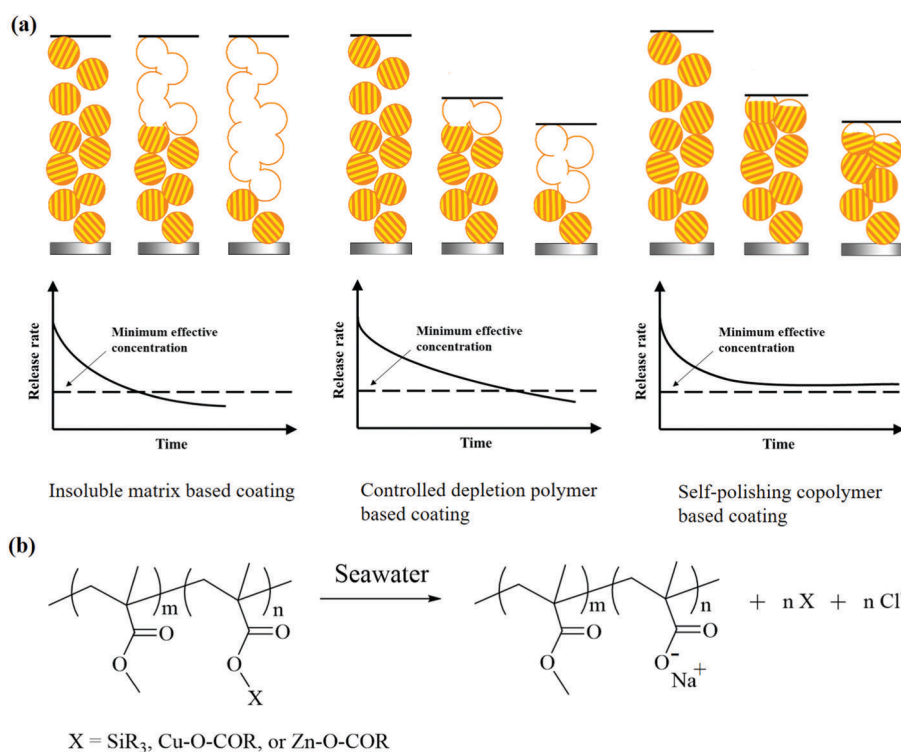


Fig. 2 (a) The schematic of coatings based on antifoulant and different polymer binders and typical release rate of antifoulant with immersion time; (b) the chemical structure and hydrolysis reaction of self-polishing copolymers.

polymers (CDP) and self-polishing copolymers (SPC). For the insoluble binders like chlorinated rubber, the release of antifoulant is only a diffusion process, which leads to empty pores in the upper layer and hinders continuous release. Consequently, the release rate decreases with time. CDP with a high content of rosin (a water-soluble natural resin extracted from trees) or rosin derivatives have a more durable release of antifoulant because they can dissolve in seawater. In comparison, SPC based on silyl, copper, or zinc acrylates are a better carrier for antifoulants. Their silyl/copper/zinc ester side groups can undergo hydrolysis to generate a hydrophilic surface, which is then polished away by water flow and antifoulants are released alongside.^{18,19} However, their surface renewal is not quick enough without strong water flow, so most SPC based coatings have poor resistance to macro- and micro-fouling organisms under static conditions when the fouling pressure is high. Moreover, slow surface erosion and continuous water absorption may cause the swelling of coatings, which negatively impacts the mechanical and antifouling performance.

To improve the performance of SPC, Bressy *et al.*^{20–24} have developed diblock copolymers of *tert*-butyldimethylsilyl methacrylate (TBDMSiMA) and methyl methacrylate (MMA) with controlled microstructures by reversible addition-fragmentation chain transfer (RAFT) polymerization for antifouling, which demonstrate a better controlled erosion rate and antifouling efficiency than the statistical copolymer coatings.²¹ They also prepared block copolymers of *tert*-butyldimethylsilyl methacrylate (MASi) and poly(dimethylsiloxane) (PDMS) by RAFT polymerization.²⁵ Besides the hydrolyzable property, the copolymers show hydrophobicity owing to the PDMS segment, and the coating with the highest receding water contact angle displays resistance to hydrophilic marine bacteria and good antifouling performance in marine field tests.

Besides the polymer binder, the antifoulant is also critical for SPC based paints. Cuprous oxide is now the most commonly used antifoulant because it can generate Cu^{2+} that denatures the enzymes and proteins of fouling organisms. It has high efficiency against hard fouling like barnacles but not to algae,²⁶ so it is usually combined with other antifoulants such as copper pyrrhione (CuPT) and zinc ethylene bisdithiocarbamate (Zineb) to have a broad-spectrum effect. Moreover, cuprous oxide accumulates in the marine environment and affects the growth of non-target organisms.²⁷ Thus, the use of organic antifoulants

to prepare copper-free antifouling coatings is more attractive because they have smaller impact on the marine environment, and they can inhibit galvanic corrosion on an aluminum substrate. However, the reported organic antifoulants are not as broad-spectrum and efficient as metal based ones.

In the future, the inevitable direction is to develop high-efficiency, broad-spectrum and low-toxicity antifoulants and polymer binders with optimized chemical structures to lower the dosage of antifoulants, so that negative ecological impacts can be minimized.

2.2 Fouling release coatings

With the growing concerns over toxic antifouling coatings, much effort is now focused on the development of fouling release coatings (FRCs) because they are eco-friendly antifouling materials through a non-stick mechanism (Fig. 3). They are capable of easily removing adhered organisms from the surface by water flow or mechanical cleaning, without releasing antifoulant. FRCs are made of silicone or fluorinated polymers with low surface energy which is essential for low adhesion strength to fouling organisms.²⁸ Since a low elastic modulus is another key factor for good fouling release ability,²⁹ silicone based FRCs generally outperform fluorinated ones and have more applications. FRCs do not undergo erosion during service because they are chemically stable in seawater. In principle, they can provide a longer antifouling duration than SPC based paints. They are also able to reduce the frictional resistance of vessels thanks to their smooth surface. However, silicone coatings have poor adhesion to the substrate (usually epoxy primer) due to their non-polar siloxane chains. In application, an additional tie-coat with silyl and polar moieties is required to improve the adhesion strength, which increases the operating time and cost. Silicone coatings are also prone to mechanical damage such as tearing and puncturing. Moreover, they have limited antifouling efficiency during idle periods because the fouling release performance requires the external shear force of strong water flow.

Many studies have been done to improve the mechanical and antifouling properties of silicone based FRCs, including incorporation of nanoparticles,^{30,31} urea/urethane bonds,^{32,33} chemically grafted antifoulants,^{34,35} amphiphiles^{36–38} and zwitterions.³⁹ For instance, we synthesized polyurea with PDMS as the soft segment, which exhibits not only excellent fouling

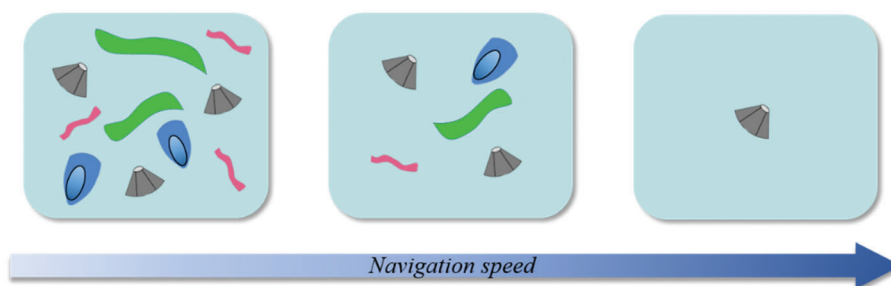


Fig. 3 The schematic of fouling release coatings.

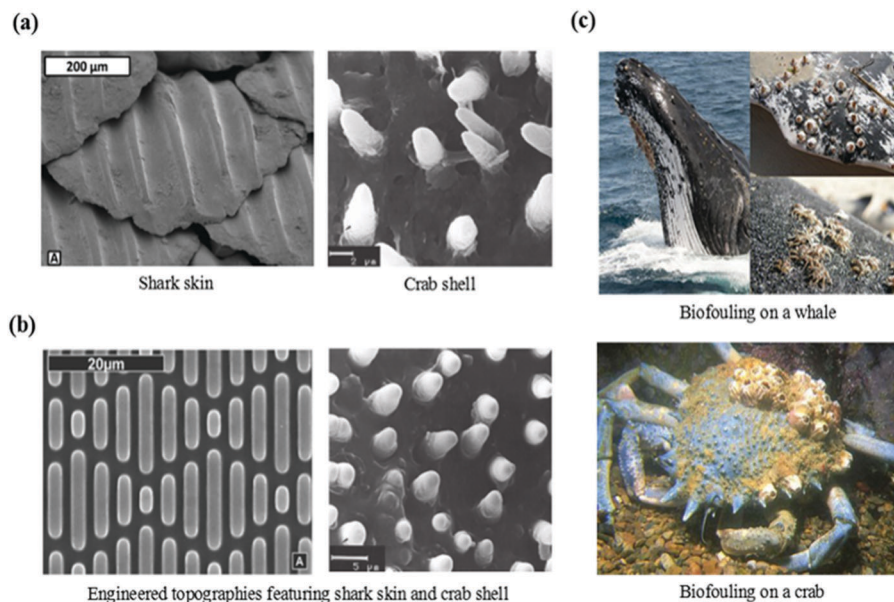


Fig. 4 (a) The topography of shark skin of *Carcharhinusbrevipinna* and crab shell of *C. pagurus*; (b) engineered surfaces with microstructures mimicking shark skin and crab shell; (c) examples of animals with microtopographical surfaces but fouled by marine organisms. Reproduced with permission from ref. 43, 46 and 53. Copyright 2004, 2007 and 2011 Taylor & Francis Group.

release properties but also significantly improved adhesion strength to the substrate compared to traditional silicone.⁴⁰ Particularly, the polymer coating is able to repair itself quickly at room temperature after damage, in air or seawater.⁴¹ This excellent self-healing property is attributed to the dynamic hydrogen bonding within the urea moieties. Cavas *et al.*³¹ reported the addition of multiwall carbon nanotubes and graphene oxide to PDMS elastomer, which improves the tensile strength and elongation without influencing the fouling release properties. To enhance the antifouling ability, Grunlan *et al.*⁴² modified PDMS elastomer with chemically grafted amphiphiles consisting of poly(ethylene glycol) (PEG) and silicone segments. The presence of amphiphilic additives improves the fouling resistance of PDMS elastomer to protein and marine micro-organisms such as bacteria and diatoms. We previously grafted antifoulant *N*-(2,4,6-trichlorophenyl)maleimide (TCM) to PDMS based polyurethane,³⁵ which shows strong inhibition of bacteria, diatoms and barnacle cyprids. The combination of low surface energy and grafted antifoulant exerts excellent antifouling ability in natural seawater for 110 days. Actually, there are many more examples of modifications of FRCs, which have been summarized in several published reviews.^{1,2} It can be anticipated that FRCs will gain even more attention in the near future, thanks to the non-toxic mechanism.

2.3 Biomimetic antifouling materials

In recent years, biomimetic and bio-inspired materials have been designed for antifouling because nature has created many marine animals free of fouling organisms, which is attributed to microtopographical surfaces, secreted bioactive molecules, mucus secretions and sloughing surface layers.^{43,44} Since sharks, whales, dolphins and crabs with microtopographical structures

on their skins or shell usually have clean surfaces with little micro- and macrobiofouling (Fig. 4), it was hypothesized that the length scale of surface textures was smaller than most fouling organisms so fewer contact points are available for adhesion.⁴⁵ To mimic such surfaces, physical methods such as photolithography, laser ablation and moulding as well as chemical synthesis have been developed.^{46–53}

For example, PDMS elastomer with engineered topographies similar to shark skin (sharklet AF™) through a photolithography treatment can reduce settlement of green alga *Ulva* spores in laboratorial bioassays.⁴⁶ Layer-by-layer spray-coating deposition of polyelectrolytes was also reported to prepare microstructured surfaces, where the sample with morphology features similar to pilot whale skin has the lowest settlement of *Ulva* spores.⁵² Yet, there are few reports on their long term antifouling ability in natural seawater. Moreover, it is recognized that the antifouling ability decreases due to age, injury or diseases.⁵⁴ Probably, secretion of bioactive molecules and mucus should be attached greater importance for naturally fouling resistant animals.

Many marine animals and plants defend themselves by generating bioactive molecules. Inspired by this phenomenon, people have extracted a large amount of natural compounds such as diterpenoids, camptothecin, capsaicin and piperine from marine bacteria, algae and invertebrates as well as terrestrial plants.^{55,56} Natural compounds are a promising source for antifoulants because they originate from the environment, making them compatible with the biological system and more specific against fouling organisms than metal ones. For example, based on the strategy of “from the sea, for the sea”, Qian *et al.*^{57,58} discovered an eco-friendly antifoulant 5-octylfuran-2(5*H*)-one (known as butenolide) derived from marine bacteria *Streptomyces sp.* Butenolide can block the adhesion-related signals and metabolism

pathway so it has strong antifouling activity. It also shows fast degradation and low toxicity against non-target organisms.⁵⁹

Enzymes secreted by animals (e.g. pilot whales) hydrolyze proteins, which are the bio-adhesives of fouling organisms such as barnacles and mussels, so enzymes can be alternative antifouling compounds.⁴¹ Peres *et al.*⁶⁰ added papain (a biodegradable proteolytic enzyme extracted from latex of papaya fruit) to rosin based binder, and the resulting coatings show good antifouling properties in marine field tests. Tremel *et al.*⁶¹ prepared a silicone coating incorporating vanadium pentoxide nanowires which mimicked the natural enzymes vanadium haloperoxidases. In this system, singlet molecular oxygen ($^1\text{O}_2$) is formed and accounted for its antifouling ability in seawater. However, a broad-spectrum antifouling effect may require the combination of several enzymes, since one enzyme is specific to certain proteins. In addition, it is difficult to retain its activity in organic solvents and dry coatings. More importantly, a thorough, time-consuming evaluation process is mandatory for both natural antifoulants and enzymes to assess their mechanisms, long-term efficiency and ecological impacts. According to regulations of the European Union and some other areas, they must be registered before use, the same as commonly used antifoulants. There are still explorations to be done on them.

Mimicking the secreted lubricating mucus from animals such as dolphins and fish provides another direction. That is, fabricating lubricating or slippery surfaces to reduce biofouling. Webster *et al.*⁶² incorporated phenylmethyl silicone oils into siloxane-polyurethane coatings with good mechanical properties, and the coatings demonstrate enhanced fouling release ability toward macroalga *U. linza*, barnacles *A. amphitrite* and mussels *G. demissa* compared to the sample without additive oil, depending on the phenylmethyl content and viscosity of oil. The coating also has excellent performance in marine field tests with water jet treatment. Aizenberg *et al.*⁶³ infused silicone oil in a cross-linked PDMS network, and the resulting surface exhibits a low attachment number and adhesive strength of mussels in laboratorial bioassay as well as good antifouling performance for 16 weeks in marine field tests. It is explained that the lubricating oil can deceive the mechanosensing ability of mussels, deter secretion of adhesive threads and decrease the molecular work of adhesion. Actually, many other materials have been designed to imitate the nepenthes pitcher plant, a tube-shaped plant lined with a slippery surface created by locking water or nectar into its microscale textures, which helps to capture insects inside. These slippery liquid-infused porous surfaces (SLIPS) exhibited excellent fouling repellency to water, oils, blood and marine organisms.^{64–66} However, these lubricating oils are constantly released in seawater, and their content in the binder is usually not high to ensure compatibility and mechanical strength. So, when the surface is 'dried out' of lubricating oils, its antifouling efficiency may decline. In addition, the influence of lubricating oils on the marine ecology still needs to be evaluated.

Enlightened by the sloughing surface of pilot whales, Ganguli *et al.*⁶⁷ used a viscoelastic surfactant and polyacrylamide-based emulsion to fabricate skin that could dissolve in seawater with time and remove the attached organisms alongside.

The diffusers protected with sacrificial skin demonstrated remarkably reduced adhesion of marine bacteria *Pseudoalteromonas* in comparison with the unprotected one. However, such a sacrificial surface dissolves in water in one week due to its highly hydrophilic nature. Actually, biological surfaces also clean themselves by active deformation and motion. Zhao *et al.*^{68,69} prepared a bio-inspired elastomeric surface with deformation stimulated by an external electric voltage, mechanical stretching, or pressured air. The surfaces with large principal strain show significantly higher removal of *Cobetia marina* biofilm and barnacles.

In brief, biomimicry has promising prospects for anti-biofouling as an eco-friendly technology. However, it is unlikely that mimicking a single biological characteristic of animals or plants can generate a broad-spectrum antifouling material with long-term efficiency. Combining multiple mechanisms such as sloughing surfaces and secretion of bioactive compounds may be a practical solution.

2.4 Protein resistant polymers

Since the initial process of biofouling is usually protein adsorption and many fouling organisms secrete protein-based bio-adhesives to settle on surfaces,^{70,71} protein-resistant materials are expected to reduce fouling accumulation. Zwitterionic polymers and PEG are well-known for their remarkable protein resistance due to the hydrogen bonding or electrostatically induced hydration layer, which forms a physical barrier to proteins.^{72–74} Accordingly, the possibility of their use in marine antifouling has been explored.^{75–79} Jiang *et al.*⁸⁰ grafted poly(sulfobetaine methacrylate) (PSBMA) brushes onto glass surfaces using surface-initiated atom transfer radical polymerization (ATRP). The zwitterionic surface displays low adhesion density of *Ulva* spores and diatom *Navicula*. Similar to PSBMA, poly(carboxybetaine methacrylate) (PCBMA) can also prevent the settlement of organisms such as barnacle *Amphibalanus Amphitrite* cyprids.⁸¹

Kang *et al.*⁸² immobilized PEG on a tannic acid coated surface, where protein adsorption (fibrinogen) and the diatom density (*Amphora coffeaeformis*) are lower than the non-treated control. Self-assembled monolayers of PEG has good resistance to the adhesion of *Ulva* spores and *Navicula* cells as well.^{83,84} PEG can also be used to modify multi-walled carbon nanotubes, which are then added to polysulfone membranes and provide resistance to bovine serum albumin (BSA).⁸⁵ In addition, PEG can be used to prepare hydrogels with a highly elastic and soft hydration layer to resist biofouling.^{86–90} For example, Liedberg *et al.*⁹¹ prepared a hydrogel of poly(ethyleneglycol) methacrylate (PEGMA) and 2-hydroxyethylmethacrylate (HEMA) via UV-initiated free radical polymerization. The hydrogel displays excellent resistance to a protein (fibrinogen) and marine micro-organisms (bacteria, algal zoospores, diatoms and barnacle larvae).

Although protein resistant polymers have excellent anti-fouling efficiency to model microorganisms in laboratorial assays, there are few reports about their antifouling performance in marine field tests. Moreover, their hydrophilic nature usually causes the coating to swell, which compromises the mechanical strength, thereby limiting their applications in seawater. Thus, we synthesized polyurethane containing PEG

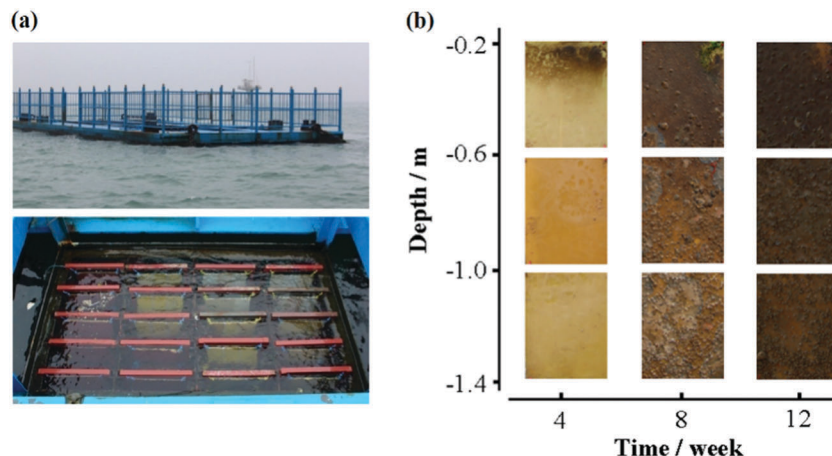


Fig. 5 (a) The raft and steel frames for marine field tests of antifouling polymer coatings; (b) coating surface of PEG-PU after immersion in natural seawater. Reproduced with permission from ref. 98. Copyright 2012 Elsevier.

or zwitterions and examined the relation between protein resistance and marine antifouling,^{92,93} because polyurethane is widely used as coatings and adhesives thanks to its remarkable adhesion to various substrates and robustness in harsh environments.^{94–97} The polyurethane with PEG as the soft segment (50 wt%) exhibits zero adsorption of fibrinogen, BSA and lysozyme, as revealed by a quartz crystal microbalance with dissipation (QCM-D).⁹² Unlike previous reports, we mainly focused on the antifouling ability in natural seawater (Fig. 5). After 4 weeks, the surface is almost free of biofouling, indicating that the protein resistant surface is effective in the early stage. However, the coating surface no longer has antifouling ability as the immersion is prolonged.

Actually, another protein resistant polymer we tested (polyurethane containing zwitterions) has a similarly short antifouling duration. After carefully inspecting the coatings after immersion, we observed some sea mud adhered to the surfaces, which formed a thin film for biological species to accumulate. Since the protein resistant surfaces are “dead”, the coverage of inorganics undermines their intrinsic antifouling ability. Therefore, a persistent protein resistant surface with surface renewability could lead to good antifouling performance in complex marine environments

because it can polish away the adhered organics and inorganics. To this end, we synthesized a copolymer of MMA and an acrylate terminated poly(ethylene oxide-co-ethylene carbonate) (PEOC) macromonomer, which shows both protein resistance and hydrolytic degradation endowed by the ethylene oxide and carbonate linkage, respectively (Fig. 6).^{98,99} The copolymer with 63 wt% PEOC effectively inhibits biofouling for more than 12 weeks in natural seawater. Compared with PEG-based polyurethane, PMMA-PEOC has hydrolytic degradation, clearly indicating the introduction of degradation can significantly improve the antifouling ability of protein resistant polymers. This was confirmed by another work of our group as well.¹⁰⁰ With complicated conditions in marine environments, degradability should be taken into account when designing PEG/zwitterion based antifouling materials.

2.5 Amphiphilic polymers

It is known that marine organisms have different preferences for the hydrophobicity of a surface. For example, diatoms tightly adhere on a hydrophobic surface while *Ulva* weakly attaches to it.¹⁰¹ A combination of hydrophobicity and hydrophilicity is expected to construct an “ambiguous” surface for

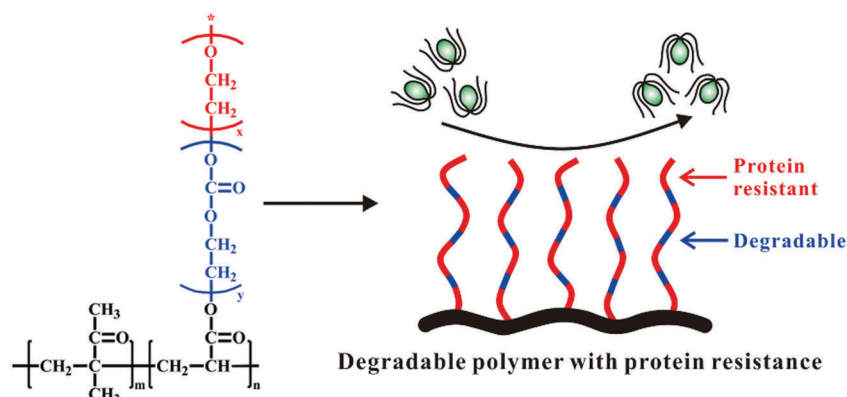


Fig. 6 The chemical structure and schematic of PMMA-PEOC copolymers. Reproduced with permission from ref. 98. Copyright 2012 Elsevier.

protein or glycoprotein adhesives secreted by the colonizing organisms and “confuses” organisms during settlement and adhesion.^{92–109}

Wooley *et al.*¹¹⁰ prepared amphiphilic networks of hyper-branched fluoropolymer and PEG on 3-aminopropyl triethoxysilane functionalized microscope glass slides. The surface has excellent resistance to BSA, lectin and lipopolysaccharide. It is also able to inhibit the settlement of *Ulva* spores and facilitate their removal, which is explained by the effect of the compositional and morphological heterogeneity. Wang *et al.*¹¹¹ prepared photo-cross-linked amphiphilic network coatings of a hydrophobic monomer dimethacryloxy-functionalized perfluoropolyether and PEG methacrylate. The coating containing PEG with a molar mass of 1100 g mol^{-1} demonstrated excellent resistance to *Ulva* spores, diatoms, and barnacle cypris larvae and a high removal rate for *Ulva* sporelings under a low-pressure water jet. Using a different approach, Ober *et al.*¹¹² synthesized a comblike block copolymer with amphiphilic side chains of ethoxylated fluoro alkyl surfactant, which was layered on top of a thick film of polystyrene based block copolymer. The amphiphilic surface shows high removal rates of both *Ulva* and *Navicula*, the latter attributed to the PEGylated surface from reconstruction of polymer chains.

However, it should be noted that almost all of these amphiphilic copolymers are composed of PEG which has good antifouling ability itself. Whether the “ambiguous” effect or PEG is the main reason is not clear yet. So far, there are few reports on the antifouling performance of amphiphilic copolymers themselves in natural seawater, but they have been chemically or physically introduced to silicone based FRCs and proven effective in enhancing antifouling ability, as discussed in the FRC section.

Generally, a variety of materials demonstrate antifouling ability but some problems are not solved yet. For the commercially available products, antifoulant releasing coatings cause negative consequences to the environment while silicone based FRCs have poor mechanical properties, and they both show declined antifouling efficiency during idle periods. Regarding bio-inspired materials, most of them are under laboratorial assessments, and they are usually difficult and costly to prepare in large quantities. For PEG/zwitterion based polymers and amphiphilic polymers, their performance in natural seawater needs to be further evaluated and improved. Therefore, exploring new strategies to combat biofouling is still challenging and urgent.

3 Dynamic surface antifouling (DSA)

Several years ago, we proposed the concept of Dynamic Surface Antifouling (DSA), where the dynamic surface refers to a changing surface that continuously renews itself in seawater and thus decreases the adhesion of biofouling. Based on this strategy, we have developed a series of degradable polymer based dynamic surfaces with tunable renewability, and excellent antifouling and mechanical performance. Such novel antifouling systems are distinct from the aforementioned materials regarding the antifouling mechanism and properties.

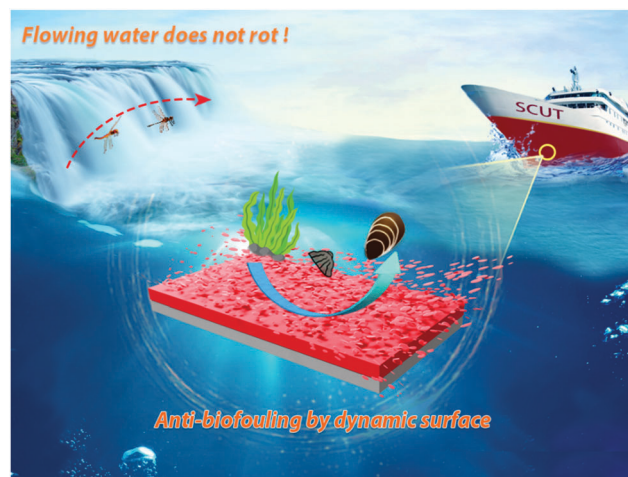


Fig. 7 The schematic of anti-biofouling by a dynamic surface constructed from a degradable polymer.

3.1 DSA mechanism

The formation of biofouling on a submerged surface is not instantaneous. It includes the landing of microorganisms, secretion of bio-adhesive and a growing process. A dynamic surface can shorten the contact time between the organism and surface, which makes their interaction close to a non-adhesive collision, thereby lowering the interaction force with fouling organisms and facilitating their escape. Actually, DSA has a meaning similar to an ancient Chinese proverb: “Flowing water does not rot” and is similar to the phenomenon that sloughing behavior of animals contributes to their fouling-free characteristic (Fig. 7).

The DSA mechanism was proven by studying the settlement of model microfoulers on surfaces of degradable polymers *via* digital holographic microscopy (DHM) which can afford the three-dimensional real-time motion behavior of microorganisms.^{113,114} The effect of the degradation rate on the near-surface motion of bacteria strains *Pseudomonas* sp. and *Escherichia coli* (*E. coli*) was investigated. Model surfaces were constructed by using δ -hexalactone (HL), an isomer of CL, as a copolymerization component to adjust the degradation rate of PCL. QCM-D tests show that the enzymatic degradation rate markedly increases with HL content. Moreover, the copolymers form surfaces with similar roughness and wettability so their effects are also eliminated. With these influences ruled out, we can extract the impact of the degradation rate on bacterial adhesion.

On a poly(CL-co-HL) surface, coverage of each bacterial strain reduces by 10–20% as the degradation rate increases. We used DHM to record and reconstruct the bacterial behavior near the surface in aquatic environments. By analyzing the trajectories of bacteria in terms of their mean square displacement (MSD), we find that when the degradation rate is higher, a larger portion of both bacteria strains demonstrates active motion rather than subdiffusive motion (Fig. 8). It is well established that the approaching bacteria exhibit subdiffusive motion on the surface mediated by flagellum or pili before irreversible adhesion.¹¹⁵ In other words, a higher degradation rate prevents the bacteria from transitioning to an irreversible state of

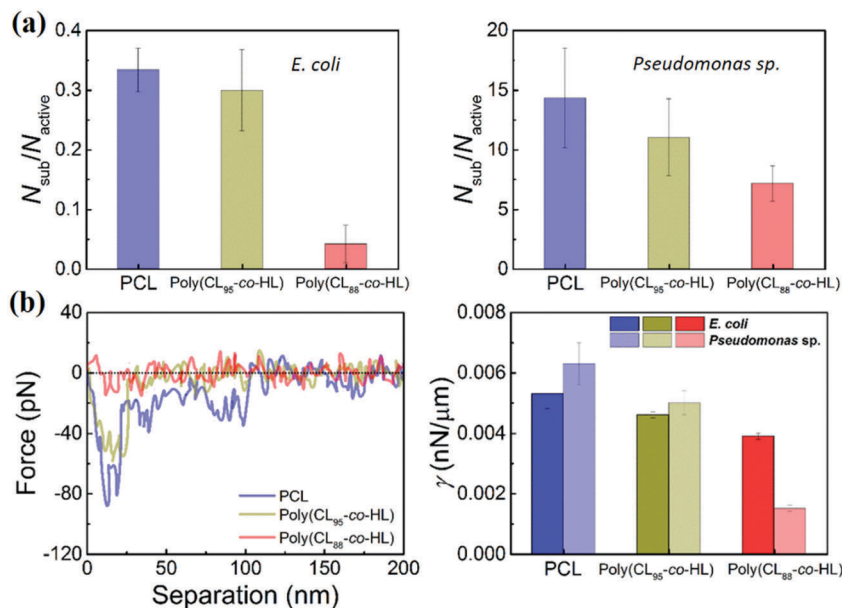


Fig. 8 (a) The ratio ($N_{\text{sub}}/N_{\text{active}}$) between the subdiffusive and active density of bacteria at $z = 0-4 \mu\text{m}$ on the surfaces; (b) typical force-separation curves of *Pseudomonas sp.* and mean normalized adhesion force (γ) of both bacteria strains on the surfaces. Reproduced with permission from ref. 113. Copyright 2017 American Chemical Society.

adhesion. We further investigated the reason behind *via* force spectroscopy measurements by atomic force microscopy (AFM). The normalized adhesion force (γ , acquired from the Gaussian fit of the force histograms) between a bacteria-covered colloidal probe and the surface decreases as the degradation rate increases. Namely, the degradation of the polymer forms a changing dynamic surface which avoids irreversible bacterial adhesion through decreased adhesion strength. These results clearly show that the dynamic surface alters the motion of bacteria in a way that favors their detachment and escape from the surface. Actually, the DSA mechanism occurs throughout the immersion in seawater because hydrolytic and enzymatic degradation always happen.

Erosion of the polymer leads to renewal of the surface. The erosion can be designated as the sum of all processes that cause mass loss of the polymer, and the processes are mainly related to the water diffusion and the water solubilization/removal of polymers by hydrolysis, ionization, protonation of a pendant group or by backbone cleavage.¹¹⁶ The erosion mechanism is generally classified into two processes: (a) surface erosion, a heterogeneous erosion process occurring on the surface of the coating with possibly linear mass loss kinetics and (b) bulk erosion where the process occurs throughout the overall thickness of the coating, leading to non-linear kinetics of mass loss. The two erosions result in the renewal of the coating surface.

A DSA coating is defined as a “coating that has a continuously changing surface in seawater.” Degradable polymers can undergo cleavage of the main chain in contact with water. This process occurs either on the surface or in the bulk, each leading to the transition from macromolecules to oligomers or small molecules which easily disperse into seawater.^{117,118}

Thus, degradable polymers are typical DSA coatings. For erosion of SPC, the first step is the hydrolysis of side groups in contact with water, and the second step is the detachment of the generated hydrophilic upper layer by seawater motion under dynamic conditions, depending on the chemical composition of co-monomers. Considering that the early stage of SPC erosion occurs at a molecular level with small molecules released from side group hydrolysis and a change in the chemical composition of the surface, SPC could be claimed as DSA coatings. However, the surface renewal behaviors of SPC and degradable polymers are quite different (Fig. 9). The former is a tearing process requiring external shear force. Thus, it is difficult for SPC to achieve a linear decrease in coating thickness with time, especially during idle periods. For degradable polymers, the surface renewal is a spontaneous process where water flow is not necessary. A degradable polyester coating could have linear kinetics of surface erosion even in idle periods. Even in the cases of copolyesters where some un-degradable segments exist, *e.g.* poly(ester-co-acrylate), the polymer backbone can still turn into oligomers with significantly decreased molecular weight because the distribution of esters with a high content is generally random,¹¹⁹ so linear kinetics of erosion could be achieved. Another outcome of the difference in their erosion is the environmental impact. The eroded SPC maintains a high molar mass in seawater, which makes it a potential source for microplastic pollution. Microplastic ingested by crustaceans and fishes hinders food from passing through the intestinal tract or causes pseudo-satiation resulting in declined food intake.¹²⁰ In contrast, biodegradable polymers can undergo enzymatic and hydrolytic cleavage of backbones, yielding oligomers or small molecules which eventually turn into carbon dioxide and water after a certain period of time. Even for

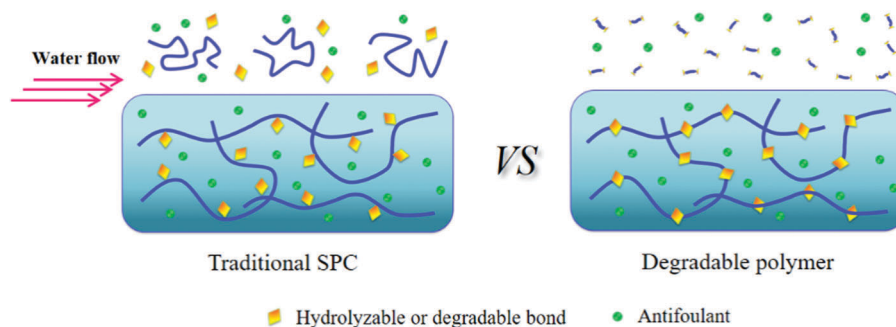


Fig. 9 The schematic of DSA coatings (SPC and degradable polymers) showing different surface renewal behaviors.

polymers containing degradable and un-degradable segments, the degraded product has a much smaller size than that of eroded SPC, so they have little negative ecological impact.

3.2 DSA systems

3.2.1 Degradable polyester based polyurethanes. Aliphatic polyesters are typical biodegradable polymers which can undergo enzymatic (from bacteria, fungi, and algae) or hydrolytic degradation *via* the cleavage of polymer main chains, leading to renewable surfaces.^{121–123} Hence, they are promising candidates for anti-biofouling. However, traditional polyesters such as poly(ϵ -caprolactone) (PCL), poly(butylene succinate) (PBS) and poly(L-lactic acid) (PLA) cannot be used directly as antifouling coatings because they have poor film forming properties and low adhesion strength to substrates.^{124,125} Moreover, their degradation in seawater is quite slow due to high crystallinity.

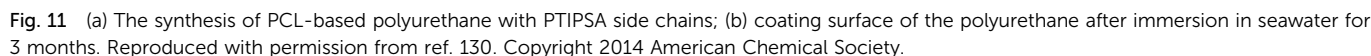
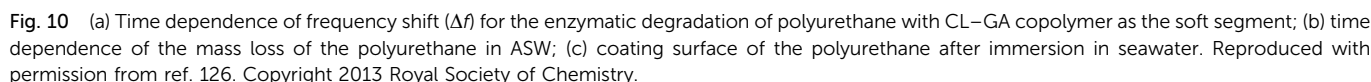
We synthesized polyurethane with PCL as the soft segment (PCL-PU) and lysine ethyl ester diisocyanate (LDI) and 1,4-butanediol as hard segments.¹²⁶ The presence of urethane moieties provides good film forming properties and adhesion to substrates. Yet, PCL-PU is still highly crystalline so its degradation rate is low. To lower the crystallinity, we prepared a dihydroxyl-terminated polyester copolymer of caprolactone (CL) and glycolide (GA) by ring-opening polymerization and used it as the soft segment of polyurethane. Differential scanning calorimetry (DSC) tests prove that the soft segment exhibits crystallinity lower than PCL and PGA because the regularity of polymer chains is destroyed by the copolymerization and hence the crystallization is hindered. Its spherulite size of crystals also notably decreases. As a result, the amorphous area increases, making it easier for water or enzyme molecules to attack the ester bonds in the polymer. The polyurethane based on CL-GA copolymer demonstrates higher enzymatic and hydrolytic degradation rates than PCL-PU and the rates increase with GA content (Fig. 10). In particular, the polyurethane with 10 mol% GA in the soft segment has good antifouling performance in seawater for 3 months without any antifoulant or protein resistant moieties. In other words, the degradable polymer itself is effective in antifouling.

Diols of poly(ethylene adipate) (PEA), poly(1,4-butylen adipate) (PBA) and poly(1,6-hexamethylene adipate) (PHA) with different densities of ester bonds were also introduced into

polyurethane as soft segments to regulate the surface renewal rate.¹²⁷ Our studies show that the mass loss of PHA-PU, PBA-PU and PEA-PU after immersion in ASW for 2 months is about 0.40, 0.75 and 0.80 mg cm⁻², respectively. That is, the mass loss increases with the ester density which determines the number of spots available for main-chain cleavage, indicating that the degradability can be readily regulated. Moreover, these polyurethane coatings have excellent adhesion strength to substrates over 4 MPa because the urethane moieties strongly interact with the polar substrate thereby forming hydrogen bonding, which is beneficial for their long-term application in marine environments. Following this design, there have been more attempts with polyester diols such as star-shaped PCL and polypropylene glycol-CL-LA copolymer, where the polyurethane also demonstrates degradability.^{128,129}

Incorporation of hydrolyzable groups into polyurethane can significantly affect the surface renewal as well. As mentioned above, silylacrylate based polymers are typical hydrolysable SPC. They can be introduced to degradable polyurethanes as side chains/groups (Fig. 11). We applied telomerization to synthesize a single-ended dihydroxyl functionalized oligomer of triisopropylsilyl acrylate (TIPSA).^{130,131} This oligomer was grafted to PCL-based polyurethane to obtain a polymer bearing degradable main chains and hydrolyzable side chains. In other words, both the backbone and side chains can give rise to the surface renewal of the polymer coating. The presence of PTIPSA profoundly influences the mass loss rate in ASW, which gradually increases from 0.10 to 0.35 wt% day⁻¹ as the TIPSA content increases from 0 to 40 wt%. Tracing the surface morphology of the polymer coating in ASW with scanning electron microscopy (SEM), we know that the surface roughness slightly varies with immersion time. This indicates that the polymer erodes with the main-chain degradation and side-chain hydrolysis, forming a continuously renewed dynamic surface. With the dual drive for surface renewal, the polymer coating containing 40 wt% PTIPSA (PU-S40) has excellent antifouling efficiency for 3 months in seawater.

More studies were conducted on the novel degradable and hydrolyzable polyurethane to reveal the influence of molecular structure. When the length of PTIPSA side chains is fixed, the mass loss shows a tendency PCL-PU < PEA-PU < PLA-PU, which is consistent with the order of main-chain ester density (PCL-PU < PEA-PU < PLA-PU). Bearing the same polyester soft



the TCM content is above 20 wt%. Accordingly, the presence of TCM accelerates the enzymatic degradation of the polymer in ASW. Meanwhile, TCM enhances the fouling resistance of PCL-PU to marine organisms. With both surface renewal and antifouling activity, the polymer coating demonstrates excellent antifouling ability in marine field tests. In another study of ours, 2-(dimethylamino)ethyl methacrylate (DEM) was grafted to PCL-PU to regulate the crystallinity, which decreases from 31.6% to 6.2% as the DEM content increases from 0 to 25 wt%.¹³³ Additionally, the presence of DEM provides anti-adhesion performance against marine bacteria *Micrococcus luteus* due to its hydrophilic nature and fully deprotonated state in seawater. Clearly, the self-renewal rate of polyurethane based dynamic surfaces can be readily tuned by the structures of soft

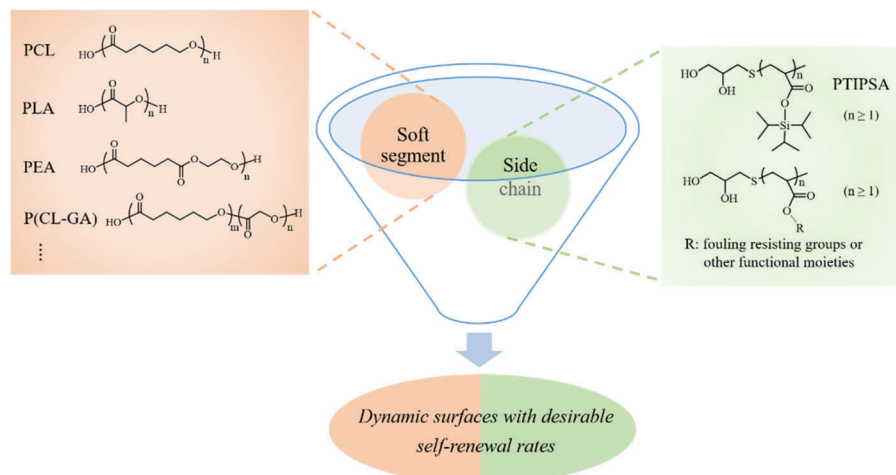


Fig. 12 The molecular design of polyurethane with different polyester soft segments and functional side chains for fabricating dynamic surfaces.

segments and side chains to fulfil the demands for various occasions (Fig. 12).

Note that the antifouling ability of dynamic surfaces increases with the degradation rate. However, every coin has two sides. Different from the reproducible skins of marine animals by their metabolism, the degradable coating has a limited thickness. A high degradation rate results in an inevitably short antifouling duration. In the biomedical field, biodegradable polymers have been used as carriers of drugs because their erosion leads to the controlled release of drugs in targeted tissues and cells.^{134,135} This approach is also applicable to marine anti-biofouling. That is, an environment-friendly antifoulant can be combined with a dynamic surface constructed from a degradable polymer with a moderate erosion rate to establish a dual-functionality system, so that durable antifouling performance is realized. In traditional polymer/antifoulant systems, the antifoulant usually has a quick release at the initial stage of immersion but the release rate drastically decreases with time, which causes waste at the beginning but declined efficacy in the late period. Optimized antifouling performance requires that the antifoulant is released at a low, stable but effective concentration and the polymer binder plays a key role.

Our studies show that degradable polyurethane is a good binder for antifoulants because its constant surface renewal in seawater releases out the antifoulant in a stable manner. We investigated the release of 4,5-dichloro-2-octyl-isothiazolone (DCOIT, a relatively eco-friendly organic antifoulant) from PEA/PBA/PHA based polyurethane by its extraction from ASW and subsequent UV spectrophotometer analysis.¹²⁷ The cumulative release of DCOIT increases linearly with the immersion time with a correlation coefficient >0.99 , indicating a stable and persistent release process (Fig. 13). As a result of surface renewal and antifoulant release, the coating has excellent antifouling performance after immersion in seawater for 6 months. From another polymer, the main-chain degradable and side-chain hydrolyzable polyurethane,¹³¹ the DCOIT release rate is also stable with immersion time. Moreover, the release rate increases with the mass loss rate of the polymer, indicating that the

degradable polymer is capable of controlling the release of antifoulant by its surface renewal rate.

We further developed eco-friendly antifouling systems constructed from degradable polymers and natural antifoulants to minimize the possible ecological impact. As mentioned above, butenolide is a natural compound with high antifouling efficiency but low toxicity to non-target marine organisms. As a starting point, we used PCL-PU80 containing 80 wt% PCL to carry and release butenolide.^{136,137} By comparing the mass loss of PCL-PU80 in natural seawater (NSW) and ASW, we know that the polymer degrades significantly faster in NSW, so the presence of enzymes secreted by bacteria and fungi accelerates the degradation (Fig. 14). We verified this by QCM-D tests, where the degradation rate increases with enzyme concentration. Namely, when biofouling is severer, more enzymes are secreted and the biodegradable polymer can have a correspondingly higher surface renewal rate. The PCL-PU80/butenolide coating exhibits excellent anti-biofilm performance in seawater when the butenolide content is ≥ 5 wt%. Yet, the coating is fouled by organisms such as barnacles and tubeworms after 3 months because the surface renewal is slow and the release rate of butenolide gradually decreases with time. Adding rosin to PCL-PU80 accelerates the polymer erosion and makes the release rate of butenolide markedly more stable. Consequently, the coating of PCL-PU80, rosin and butenolide (5 wt%) shows better antifouling ability than the samples without rosin, including the one of PCL-PU and butenolide (10 wt%) with a higher release rate. This clearly proved that both the surface renewal and release of antifoulant inhibit biofouling.

Interestingly, the release of butenolide is temperature-responsive. As the temperature increased from 20 to 30 °C, the rate after 4 weeks increases from ~ 7.5 to $\sim 13.5 \mu\text{g day}^{-1} \text{cm}^{-2}$ because it has a melting point of 23 °C and hence a higher mobility in the polymer at 30 °C. As we know, biofouling accumulation is faster when the temperature is higher. Namely, the system is capable of releasing more butenolide when needed. Being both enzyme and temperature responsive, the combination of biodegradable polymers and butenolide is a

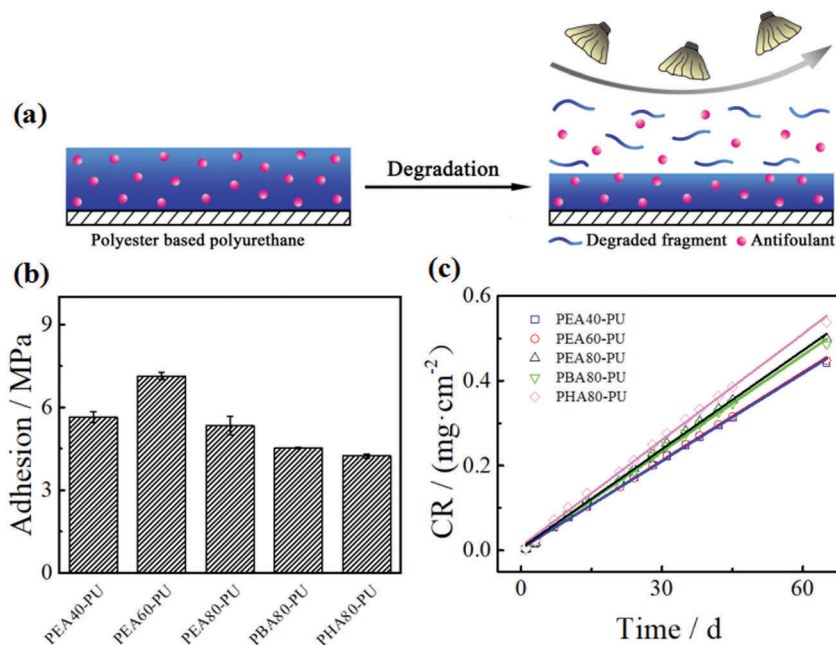


Fig. 13 (a) The schematic of the dynamic surface of PEA/PBA/PHA based polyurethane which controls the release of an eco-friendly antifoulant; (b) the adhesion of the coatings to epoxy substrate; (c) the cumulative release of antifoulant from the coatings. Reproduced with permission from ref. 127. Copyright 2017 Elsevier.

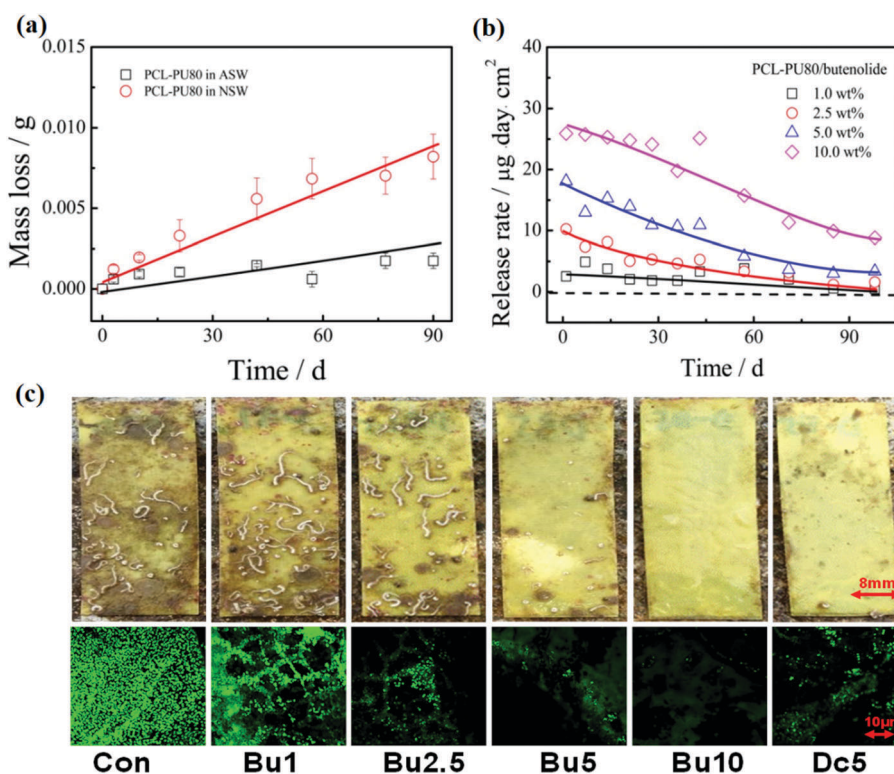


Fig. 14 (a) The mass loss of PCL-PU80 in natural or artificial seawater; (b) the release rate of butenolide from PCL-PU80; (c) the PCL-PU80/butenolide/DCOIT surface after immersion in seawater for 36 days and the fluorescence images of biofilms developed on the coatings. Reproduced with permission from ref. 136 and 137. Copyright 2017 American Chemical Society and Copyright 2018 Taylor & Francis Group.

smart environment-adaptive system. We also explored the combination of degradable polyurethane with natural compounds of fungal isolates from deep-sea sediments in the South

China Sea. The coating of PCL-PU and extract of *Aspergillus westerdijkiae* displays strong antifouling activity in seawater after 4 months.¹³⁸

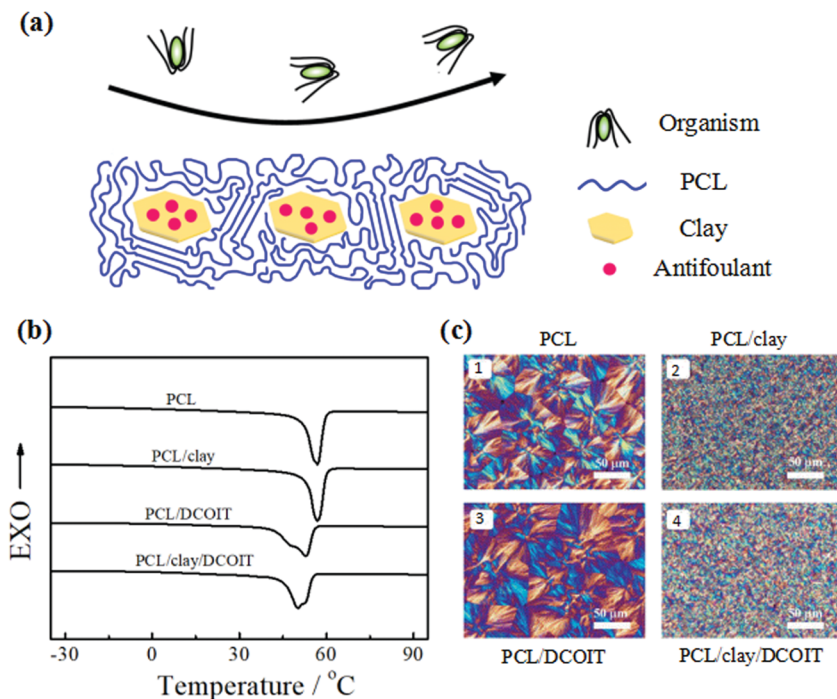


Fig. 15 (a) The schematic of the PCL/clay/DCOIT composite; (b) DSC curves of PCL and the composite; (c) the POM images of PCL and the composites. Reproduced with permission from ref. 139. Copyright 2014 Royal Society of Chemistry.

3.2.2 Degradable modified polyesters. Modification of polyester to reduce its crystallinity or crystal size can also lead to dynamic surfaces for anti-biofouling. By a facile solution mixing process, we prepared a PCL/clay (Kaolin, $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$) composite coating.¹³⁹ The crystallinity of the composite is close to pure PCL, as revealed by DSC (Fig. 15), so the mechanical strength is maintained. Meanwhile, energy dispersive spectroscopy (EDS) proves the presence of clay in the cross section of the composite, which remarkably decreases the spherulite size of PCL, demonstrated by polarizing optical microscopy (POM). The smaller spherulites increase the amorphous area and facilitate the penetration of water and enzyme molecules. Consequently, PCL/clay exhibits significantly faster enzymatic and hydrolytic degradation than pure PCL. DCOIT was added to the composite, from which it is gradually released with the polymer degradation. The coating was almost free of biofouling after immersion in seawater for 10 months, thanks to the joint effect of the dynamic surface and released antifoulant.

Blending biodegradable polymers together (at least one being degradable) can achieve the same goal. We prepared a PCL/PBS composite with good film forming properties simply by mixing them in solution.¹⁴⁰ Although both PCL and PBS homopolymers have a large spherulite size, their blend demonstrates notably smaller spherulites because PBS crystals cause surrounding confinement to PCL crystals and thus inhibit their growth. With increased amorphous interfacial area, the PCL/PBS (70/30, w/w) composite coating has a mass loss of $\sim 0.5 \text{ mg cm}^{-2}$ after immersion in ASW for 4 months while the one of pure PCL is $\sim 0.3 \text{ mg cm}^{-2}$. The composite is also used as a carrier for DCOIT, which has a linear cumulative

release in ASW with a correlation coefficient > 0.99 (Fig. 16), thanks to the uniform dispersion of DCOIT in the PCL/PBS blend before and after immersion in ASW for 4 months as revealed by EDS. As the polymeric surface constantly erodes, the antifoulant is released stably alongside. Moreover, the release rate increases with the degradation rate of polymers, showing a well-controlled tunable release behavior. Nishida *et al.*¹⁴¹ also prepared polymer tubes from blends of linear low-density polyethylene (LLDPE) and PLA (80/20, w/w) by an extrusion molding method. Such an LLDPE-PLA tube shows lower settlement of barnacles and mussels than the LLDPE control after immersion in seawater for 173 days. Yu¹⁴² reported a poly (3-hydroxy-butyrate) (PHB) and PCL composite for anti-biofouling, and the highest degradation rate is obtained at a weight ratio of 75/25 (PHB/PCL).

Polyester can be chemically modified for anti-biofouling. Faÿ *et al.*¹⁴³ synthesized poly(CL-*block*-anhydride) by a melt condensation reaction. The copolymer with hydrolytic degradability was used as a carrier of antifoulant (Diuron). The resulting coating was able to inhibit the adhesion of bacteria and microalgae in marine field tests. However, the coating completely erodes after immersion for 10 weeks due to the presence of the anhydride block which is known to degrade fast. They also copolymerized CL with LA or δ -valerolactone (VL) by ring opening polymerization with tetrabutoxytitanium as a catalyst. This poly(ester-*co*-ester) shows crystallinity lower than that of each homopolymer.¹⁴⁴ The copolymers were tested as binders for antifoulants (cuprous oxide and Diuron) but the antifouling ability in seawater was not reported. Afterwards, they synthesized triblock copolymer PCL-PDMS-PCL with degradability and

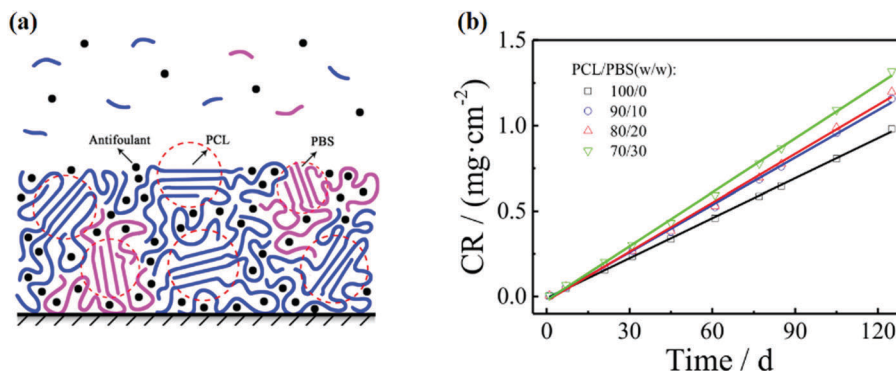


Fig. 16 (a) The schematic of the PCL/PBS composite and antifoulant; (b) the cumulative release of DCOIT from the PCL/PBS composite. Reproduced with permission from ref. 140. Copyright 2016 Elsevier.

hydrophobicity, where the latter comes from the PDMS segments.¹⁴⁵ The copolymer was mixed with antifoulants (dichlofluanide, zinc pyrithione and copper thiocyanate) to prepare coatings. The coatings based on copolymers containing 37 and 51 wt% of PDMS has better antifouling ability against biofilm and macroalgae than pure PCL because the introduction of PDMS lowers the crystallinity and improves the degradability as well as release of antifoulants. However, the film forming property or adhesion strength of the modified polyesters obtained from physically blending or copolymerization of lactones is generally poorer than polyester based polyurethane, due to the relatively high crystallinity and lack of adhesive moieties. Cracking or detachment of the coating from the substrate could happen sometimes.

Similar to polyurethanes, polyacrylates are extensively used as adhesives and coatings due to their outstanding coating performance on various substrates, but they cannot erode in

seawater unless bearing hydrolyzable side groups such as silyl esters, as discussed above. Combining the advantages of degradable polymers and polyacrylate is expected to create a novel type of efficient antifouling material. However, monomers with different polymerizable groups have drastically different reactivities and polymerization mechanisms, making hybrid copolymerization of such monomers tremendously challenging.

In 2012, we found the anionic hybrid copolymerization where cyclic and vinyl monomers polymerize into the same chain with a metal-free organic catalyst, phosphazene base (*t*-BuP4).¹⁴⁶ Fig. 17 shows the schematic of hybrid copolymerization and some typical monomers for the reaction. This novel copolymerization reaction makes synthesis of amorphous poly(ester-*co*-acrylate) possible. By this approach, we synthesized a random copolymer of CL and MMA with a simple structure.¹⁴⁷ Afterwards, we prepared hyperbranched PCL with glycidyl methacrylate as a branching agent¹⁴⁸ and poly(CL-*co*-methacrylic acid)¹⁴⁹ where

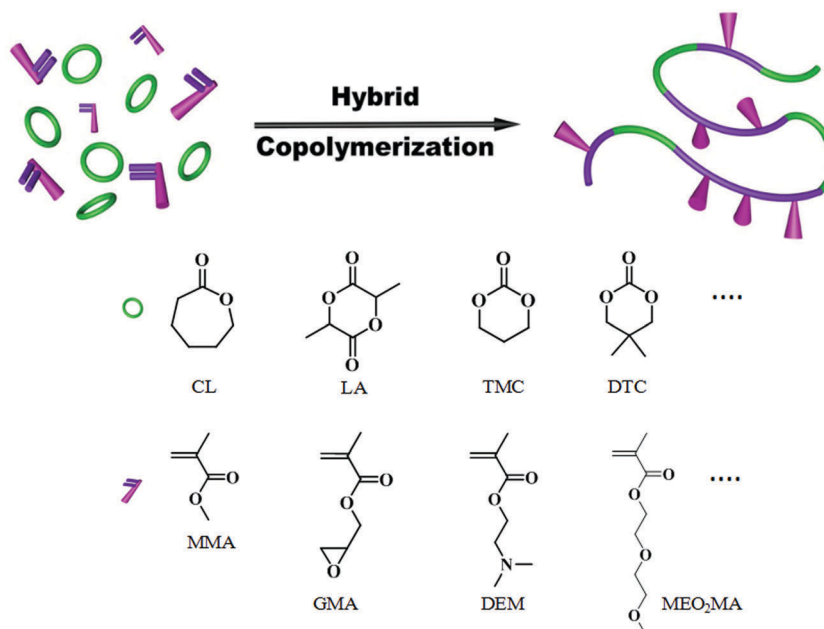


Fig. 17 The schematic of hybrid copolymerization and some typical monomers.¹⁴⁶

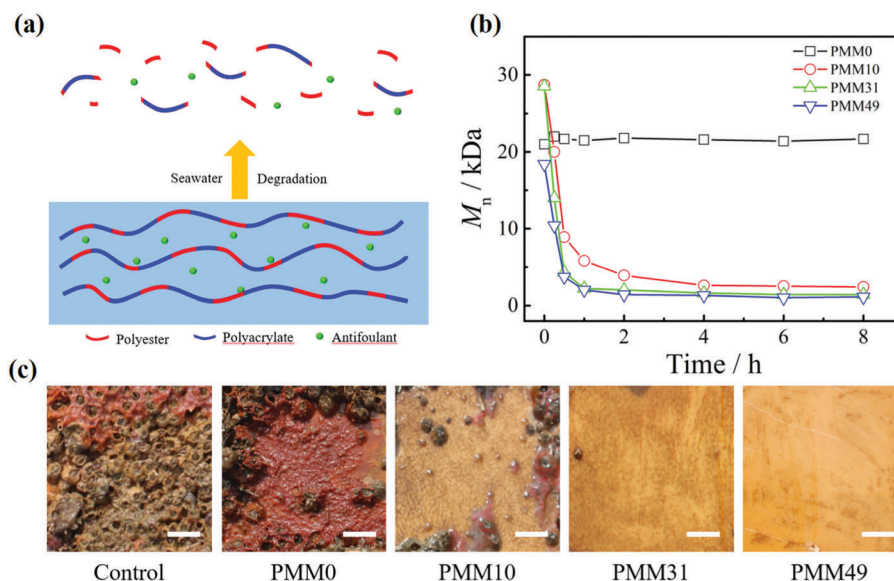


Fig. 18 (a) The schematic of the MDO-MMA copolymer; (b) time dependence of the molar mass of the MDO-MMA copolymer in 0.1 M KOH methanol solution; (c) coating surface of the copolymer and DCOIT after immersion in seawater for 3 months. Reproduced with permission from ref. 156. Copyright 2018 Elsevier.

the degradation of ester bonds is accelerated by a small amount of hydrophilic methacrylic acid. To enhance the fouling resistance of such polymers, we synthesized a copolymer of LA and 2-(2-methoxyethoxy) ethyl methacrylate (MEO₂MA) *via* hybrid copolymerization.¹⁵⁰ The copolymer has enzymatic degradation and non-specific protein resistance as well as a low cytotoxicity proven by cell viability tests.

We also exploited radical ring opening polymerization (RROP) to prepare poly(ester-*co*-acrylate). RROP is able to randomly insert ester bonds into the C-C backbone of vinyl polymers, yielding a novel type of degradable polymers.^{151–154} It opens a window to functional polyesters by copolymerizing cyclic ketene acetals (CKA) such as 2-methylene-1,3-dioxepane (MDO) and 2-methylene-4-phenyl-1,3-dioxolane (MPDO) with

diversified vinyl monomers. For example, a copolymer of MDO, oligo(ethylene glycol) methacrylate (OEGMA) and glycidyl methacrylate (GMA) has been prepared *via* RROP, which combines degradability, biocompatibility and adhesion for medical glue.¹⁵⁵ Additionally, polyester obtained from RROP usually has low crystallinity due to the branched structure, making it readily degradable. We prepared a copolymer of MDO and MMA *via* RROP.¹⁵⁶ The amorphous copolymer has excellent film forming properties with good adhesion to substrates due to the adhesive nature of MMA moieties. The main-chain esters enable the molar mass of the copolymer to decrease quickly in an alkaline solution while the one of PMMA does not change (Fig. 18). The coating is able to degrade enzymatically and hydrolytically, and both degradation rates increase with main-chain ester density.

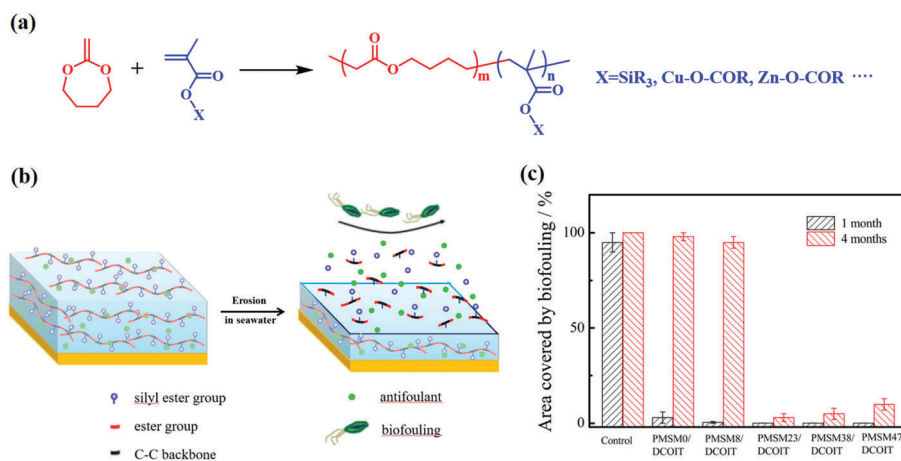


Fig. 19 (a) The synthetic route of DSPC with degradable main chains and hydrolyzable side groups; (b) the schematic of DSPC based on MDO and silyl methacrylate. (c) The percentage of area covered by biofouling for the MDO-TBSM-MMA copolymer after immersion in seawater. Reproduced with permission from ref. 158. Copyright 2015 American Chemical Society.

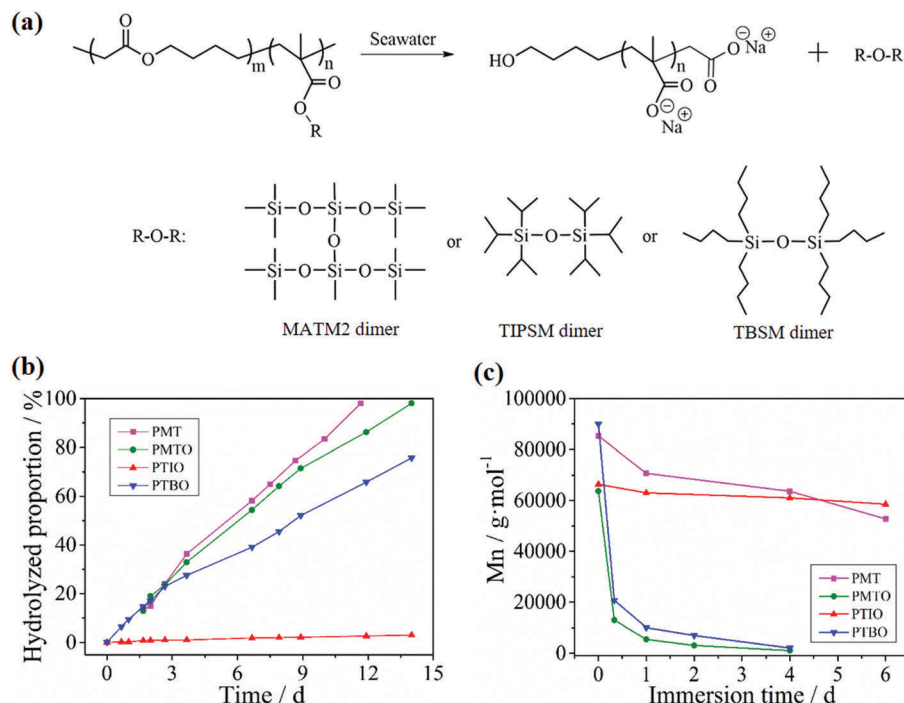


Fig. 20 (a) The degradation and hydrolysis reactions of the MDO-silyl methacrylate copolymer in seawater; (b) time dependence of the hydrolyzed molar proportion of silyl ester units in THF- d_8 with ASW at room temperature; (c) time dependence of the molar mass of the copolymer immersed in ASW at 60 °C. Reproduced with permission from ref. 159. Copyright 2018 Royal Society of Chemistry.

The copolymer can be applied as a controlled release binder for DCOIT and the coating had good antifouling ability in seawater when the copolymer contained a high content of polyester (>31 mol%). We further explored the use of acrylate monomers bearing fouling resistant moieties by RROP, obtaining an MDO-DEM copolymer with enzymatic degradation regulated by main-chain esters.¹⁵⁷ Moreover, the copolymer coating exhibits excellent antibacterial (*M. luteus* and *Pseudomonas* sp.) and protein resistant (BSA, lysozyme and fibrinogen) abilities owing to DEM.

3.2.3 Degradable and hydrolyzable poly(ester-co-acrylate)s.

Improving the surface erosion of traditional SPC binders has significance for their antifouling application in static immersion. We developed degradable SPC (*i.e.* DSPC) made of poly(ester-co-silyl methacrylate). We inserted ester bonds into the backbone of a silyl acrylate copolymer for the first time by copolymerizing MDO, tributylsilyl methacrylate (TBSM) and MMA *via* RROP (Fig. 19).¹⁵⁸ The copolymer with 23 wt% polyester exhibits a mass loss of ~13 wt% and water absorption of ~24 wt% after immersion in ASW for 90 days while the copolymer without polyester has a mass loss of ~5 wt% and water absorption of ~77 wt%. Namely, the degradable main chains can significantly improve the erosion of the TBSM based copolymer and avoid swelling, which is particularly important for low-speed occasions. Such a copolymer can be used as a binder of DCOIT and the release rate is stable in ASW. Consequently, the coating has excellent antifouling properties during static immersion in seawater while the copolymer without polyester seriously swells and has poor antifouling ability.

In the MDO-TBSM system, the ratio of polyester and silyl methacrylate is critical to optimize their properties because the main-chain degradation is slower than TBSM hydrolysis. Considering the structure of the silyl ester group may also influence the coating properties, we copolymerized MDO with bis(trimethylsiloxy)methylsilyl methacrylate (MATM2) and triisopropylsilyl methacrylate (TIPSM) in addition to TBSM, where the main-chain ester content was fixed for all three copolymers (Fig. 20).¹⁵⁹ The hydrolysis rate of silyl ester increases (TIPSM < TBSM < MATM2) as its steric hindrance decreases (TIPSM > TBSM > MATM2) and the presence of

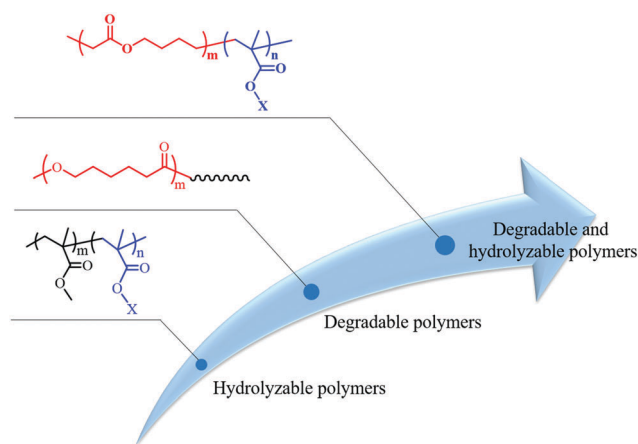


Fig. 21 The development of antifouling polymer coatings with renewable surfaces.

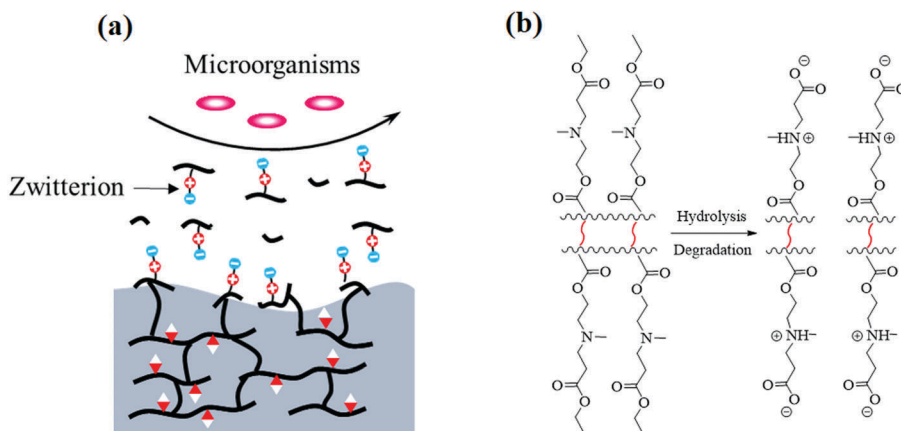


Fig. 22 (a) The schematic of the degradable polymer with hydrolysis-induced zwitterions; (b) the main-chain degradation and side-group hydrolysis of the cross-linked MDO-TCB-MAMC copolymer; reproduced with permission from ref. 160. Copyright 2018 American Chemical Society.

polyester does not show an obvious effect on the rate. MDO-MATM2 and MDO-TBSM copolymers demonstrate significant decreases in molar mass by 90% and 86% after immersion in ASW at room temperature for 7 weeks, mostly due to main chain degradation. This process could be accelerated at 60 °C. In comparison, the MDO-TIPSM copolymer did not show a notable reduction in molar mass at both temperatures. This is because when the silyl ester hydrolyzes quickly, carboxyl is generated quickly and water absorption is relatively high, both facilitating the degradation. Namely, both main-chain and side-group structures are critical for DSPC. Besides silyl acrylate based ones, DSPC from zinc and copper acrylates can also be prepared by RROP, as zinc/copper based SPC are widely used for antifouling as well. Apparently, DSPC have a bright future as a new generation of antifouling materials (Fig. 21).

We also improved the performance of DSPC using acrylate monomers containing both hydrolyzable and fouling resistant moieties. As shown in Fig. 22, we synthesized a copolymer of MDO, tertiary carboxybetaine ester (TCB) and 7-methacryloyloxy-4-methylcoumarin (MAMC), and subsequently cross-linked it under UV.¹⁶⁰ The TCB groups are able to hydrolyze in ASW, leading to a self-polishing effect whilst generating zwitterions. Unlike conventional zwitterionic polymers which are prone to swell in water, the bulk with unhydrolyzed TCB groups stays hydrophobic. Meanwhile, the hydrophilic surface with zwitterions is renewed by main-chain degradation, which is favorable to maintain both mechanical robustness and fouling resistance. The copolymer coating exhibits high adhesion strength to the substrate (>2.5 MPa) even after immersion in ASW. The coating has excellent resistance to bacteria (*Pseudomonas* sp., *Vibrio alginolyticus* and *Escherichia coli*.) and proteins (BSA, lysozyme and fibrinogen). Such a degradable polymer with hydrolysis-induced zwitterions is expected to find maritime and biomedical applications.

4 Summary and outlook

In summary, dynamic surface antifouling (DSA) is an effective and promising strategy. Based on this approach, various degradable polymers with excellent surface renewability,

mechanical properties and antifouling performance have been designed, including state-of-the-art degradable self-polishing copolymers (DSPC). Moreover, the systems of dynamic surfaces in combination with eco-friendly antifoulants have shown durable antifouling efficiency, due to constant renewal of the surface and controlled release of antifoulants.

Complexities of fouling organisms and marine environments create a tremendous challenge for anti-biofouling. Materials based on a single antifouling mechanism can hardly meet the needs of different occasions. In future studies, the design of antifouling materials should combine multiple mechanisms. Broad-spectrum fouling resistant or drag-reduction moieties can be incorporated into degradable polymers to generate multifunctional dynamic surfaces. Environment-adaptive dynamic surfaces with response to fouling pressure, temperature, salinity and navigation speed are another important direction for varying service conditions. Moreover, dynamic surfaces can be extended with other materials beyond degradable polymers. For instance, silicone based FRC with renewable surfaces can be designed by introducing hydrolyzable bonds, so that the advantages of both types of materials are merged. Last but not least, degradable polymers are not the only candidate for fabricating dynamic surfaces. That is, a constantly deforming surface may reduce biofouling as well. In this approach, highly elastic materials with surface deformation under external stimulation should be further explored in addition to the aforementioned silicone stimulated by an electric voltage, mechanical stretching, or pressured air. Undoubtedly, dynamic surfaces will play a more important role in marine anti-biofouling.

Conflicts of interest

There are no conflicts to declare.

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